Paper 1

BALNet: Balanced Attack Learning Network for Web Attack Detection via Adaptive Sampling, LDA, and CNNs (2026)

Simple Explanation

> This paper proposes a web attack detection system using a CNN model. The authors noticed that some attack types had very few samples, which reduced detection accuracy. To solve this, they used adaptive sampling to balance the dataset and LDA for feature extraction before training the CNN. The system was tested on multiple web attack categories and showed improved classification performance.

What did you learn?

> I learned that balancing the dataset is important because an imbalanced dataset can reduce model accuracy. It also showed that deep learning is effective for web attack detection.

Why is it related to your project?

> Both projects detect multiple web attacks using deep learning. However, my project uses a Character-Level Stacked BiLSTM model with Explainable AI and Threat Intelligence instead of CNN.

Guide may ask

Q: Why didn't you use CNN?

Answer:

> CNN works well for local patterns, but BiLSTM is better at understanding sequential text such as HTTP requests and URLs.

Paper 2

Swift Detection of XSS Attacks Using Hybrid Semantic Embeddings (2025)

Simple Explanation

> This paper focuses on detecting Cross-Site Scripting (XSS) attacks. The authors used hybrid semantic embeddings to better understand the meaning of web content before classifying it with AI techniques.

What did you learn?

> I learned that better feature representation improves XSS detection accuracy.

Why is it related?

> My project also detects XSS attacks, but instead of semantic embeddings, I use character-level tokenization so the model can detect obfuscated payloads.

Guide may ask

Q: Why didn't you use semantic embeddings?

Answer:

> Character-level tokenization is more suitable for web attacks because malicious payloads often contain symbols and special characters that word embeddings may not capture effectively.

Paper 3

UniEmbed: Detecting SQL Injection and XSS Using Multiple Feature Fusion (2025)

Simple Explanation

> This paper combines multiple embedding techniques such as Word2Vec, FastText, and USE to improve SQL Injection and XSS detection.

What did you learn?

> Combining different feature extraction methods can improve classification accuracy.

Relation

> My project also detects SQL Injection and XSS, but I use a single Character-Level BiLSTM model instead of combining multiple embeddings.

Guide may ask

Q: What is feature fusion?

Answer:

> Feature fusion means combining different types of features into one representation before classification.

Paper 4

Web-Based Phishing URL Detection Using Deep Learning Optimization Techniques (2025)

Simple Explanation

> This paper focuses only on phishing URL detection. The authors optimized feature extraction using Variational Autoencoders and optimization algorithms before classification.

What did you learn?

> Feature optimization improves phishing detection accuracy.

Relation

> My project also detects phishing URLs, but it additionally detects SQL Injection, XSS, Path Traversal, and Normal Traffic.

Guide may ask

Q: Does your project only detect phishing?

Answer:

> No. Phishing detection is only one module. My project performs five-class classification.

Paper 5

HawkPhish-DNN Cybersecurity Model (2025)

Simple Explanation

This paper proposes a phishing detection model using a Deep Neural Network and Harris Hawk Optimization to improve model performance.

What did you learn?

Optimization algorithms can improve deep learning accuracy.

Relation

I did not use optimization algorithms because my Character-Level Stacked BiLSTM already achieved good performance.

Guide may ask

Q: What is Harris Hawk Optimization?

Answer:

It is a nature-inspired optimization algorithm used to find better model parameters automatically.

Paper 6

Penetration Tests on Web Applications Using SQL Injection Attacks (IEEE 2025)

Simple Explanation

> This paper studies SQL Injection by performing penetration testing on web applications. It explains how attackers exploit SQL queries and how developers can identify vulnerabilities.

What did you learn?

> I understood how SQL Injection attacks work in real applications.

Relation

> My project automatically detects SQL Injection requests before they reach the application.

Guide may ask

Q: Does this paper use AI?

Answer:

> No. It mainly focuses on penetration testing and attack analysis rather than deep learning.

Paper 7

Phishing Behavior Detection via Adversarial Domain Adaptation (2024)

Simple Explanation

> This paper uses adversarial domain adaptation to improve phishing detection across different blockchain environments.

What did you learn?

> AI models can be adapted to work across different environments.

Relation

> My project detects phishing URLs in a general web environment instead of blockchain.

Guide may ask

Q: What is domain adaptation?

Answer:

> It allows a model trained on one dataset to perform well on another related dataset.

Paper 8

L-xaids: Explainable AI Framework for Intrusion Detection Systems (2025)

Simple Explanation

> This paper introduces Explainable AI using LIME to explain intrusion detection decisions. Instead of only predicting attacks, it tells users why a prediction was made.

What did you learn?

> Explainable AI increases user trust and helps security analysts understand predictions.

Relation

> My project also includes Explainable AI, but I generate simple human-readable explanations instead of using the LIME framework.

Guide may ask

Q: Why is Explainable AI important?

Answer:

> It helps users understand why the model predicted an attack instead of showing only the final result.

Paper 9

BS-GAT: Graph Neural Network Based Intrusion Detection (2025)

Simple Explanation

This paper uses a Graph Attention Network (GAT) for intrusion detection by learning relationships between connected data.

What did you learn?

Different deep learning architectures can also solve cybersecurity problems.

Relation

My project uses Stacked BiLSTM because web requests are sequential text rather than graph data.

Guide may ask

Q: Why didn't you use Graph Neural Networks?

Answer:

Graph Neural Networks are suitable for graph-structured data, while HTTP requests and URLs are sequential text, making BiLSTM a better choice.

Paper 10

Federated Learning for Intrusion Detection in IoT (2025)

Simple Explanation

> This paper proposes a privacy-preserving intrusion detection system using Federated Learning. Models are trained across multiple devices without sharing raw data.

What did you learn?

> Federated Learning protects user privacy during model training.

Relation

> My project uses centralized training because it focuses on web attack detection rather than distributed IoT devices.

Guide may ask

Q: Why didn't you use Federated Learning?

Answer:

> My project is designed for centralized web security. Federated Learning is more suitable for distributed environments such as IoT networks.

Questions That Almost Every Guide Asks

Q1. Why did you choose these papers?

Answer:

I selected recent IEEE and Springer papers because they focus on deep learning, web attack detection, phishing detection, Explainable AI, and intrusion detection, which are directly related to my project.

Q2. Which paper influenced your project the most?

Answer:

The BALNet paper influenced the deep learning approach for web attack detection, while the L-xaids paper inspired the Explainable AI module.

Q3. How is your project different from these papers?

Answer:

Most papers focus on a single problem such as SQL Injection, XSS, or phishing. My project combines five-class attack detection with Explainable AI, Threat Intelligence, a Character-Level Stacked BiLSTM model, a FastAPI-based Web Application Firewall, a Chrome Extension, and a Streamlit dashboard in one integrated framework.

Q4. Why did you choose BiLSTM?

Answer:

Because HTTP requests and URLs are sequential text. BiLSTM can understand the context from both forward and backward directions, making it effective for web attack detection.

Q5. Which paper is closest to your project?

Answer:

The BALNet paper is the closest because it also focuses on deep learning-based web attack detection. However, my project extends this idea by adding Explainable AI, Threat Intelligence, and intelligent threat mitigation.